KITABU QUEST

S5

Subsidiary Mathematics

Mascot: mountain gorilla
Scene: students solving applied mathematics problems with graphs, models, and a mountain gorilla mascot
CBC aligned draft learner book

Think • Explore • Achieve

Generated by Kitabu AI Learning Studio

Title Page

KITABU QUEST S5 Subsidiary Mathematics

Rwanda REB-CBC learner book

Mascot: mountain gorilla

This KITABU QUEST book turns the curriculum into short lessons, worked examples, guided activities, practice, review, and home links for Rwanda learners.

How To Use This Book

This book helps S5 learners study Subsidiary Mathematics one teachable topic at a time.

1. Start with the inquiry question and say what you already know.
2. Read the Learn section slowly and mark new words.
3. Try the worked example before reading the full solution.
4. Complete the activity and practice tasks.
5. Correct your answer using the success criteria.
6. Ask for support when your answer is still unclear.

Subject method: Learn through examples, activities, practice, and reflection.

Learning Skills In This Book

Each topic builds knowledge, skill, values, communication, collaboration, critical thinking, creativity, digital awareness, and self-confidence.

Study actively: speak answers aloud, draw when useful, show your working, cite evidence, use local examples, and correct mistakes after feedback.

A strong answer is specific. It names the idea, gives evidence or method, applies it to the task, and checks whether it answers the question.

Table Of Contents

Use this table to move through the book one topic at a time. Each topic has a lesson opener, clear teaching, a model or example, guided work, practice, and checks for understanding.

As you finish each topic, return to the success criteria and correct one answer before moving on.

Trigonometric functions and equations: Lesson Opener

Learning focus: Trigonometric functions and equations

Country link: a Rwanda school, market, farming, building, transport, or data problem for Trigonometric functions and equations.

By the end of this topic, you should be able to:
- solve trigonometric equations and related problems using trigonometric functions and equations

Inquiry questions:
- How can trigonometric functions and equations help you solve a real problem in Subsidiary Mathematics?

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Trigonometric functions and equations: Learn

Trigonometric functions and equations teaches a mathematical idea by using a clear model, a labelled representation, step-by-step working, and a final check.

Key words:
- Trigonometric functions and equations
- Trigonometric
- functions
- equations
- method
- model
- unit
- answer

Trigonometric functions and equations: Worked Example

Worked example for Trigonometric functions and equations: Read the task, choose a representation, solve one step at a time, and check the answer in context.

Model:
1. Name what is known.
2. Choose a diagram, table, equation, or model.
3. Complete the calculation or reasoning.
4. Check the result with estimation or inverse reasoning.

Use a Rwanda classroom, market, garden, transport, or home example.

Trigonometric functions and equations: Vocabulary And Study Skill

Use these words and study moves before you practise Trigonometric functions and equations. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Trigonometric functions and equations
- Trigonometric
- functions
- equations
- method
- model
- unit
- answer

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Trigonometric functions and equations without a clear example, method, or evidence.

Trigonometric functions and equations: Guided Activity

Guided activity for Trigonometric functions and equations:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Trigonometric functions and equations: Practice And Correction

Practice for Trigonometric functions and equations:

Main outcome: solve trigonometric equations and related problems using trigonometric functions and equations.

1. Solve one trigonometric functions and equations question using labelled steps.
2. Show your method with a drawing, table, number sentence, or diagram.
3. Check your answer and write one sentence explaining it.

Answer check:
Your answer should show a clear model, correct method, correct labels, and a final sentence.

Correction check:
- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to trigonometric functions and equations and solve it.

Trigonometric functions and equations: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Trigonometric functions and equations.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- method is clear
- working is shown step by step
- answer is checked in context

Trigonometric functions and equations: Outcome Check

Outcome check for Trigonometric functions and equations: Solve trigonometric equations and related problems using trigonometric functions and equations.

1. Write one clear example from the topic.
2. Solve it with labelled steps.
3. Check the answer in context.

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Sequences: Lesson Opener

Learning focus: Sequences

Country link: a Rwanda school, market, farming, building, transport, or data problem for Sequences.

By the end of this topic, you should be able to:
- understand, manipulate and use arithmetic, geometric sequences
- group investigation : If the bankrates increase or decrease unexpectedly, learners discuss or investigate how in the next n- years:
- the deal stays fair 30

Inquiry questions:
- How can sequences help you solve a real problem in Subsidiary Mathematics?

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Sequences: Learn

Sequences teaches how numbers change in a regular way.

Core idea:
- Look at how each term changes to the next term.
- A pattern can add, subtract, multiply, divide, or combine steps.
- Write the rule before extending the pattern.

Key words:
- Sequences
- method
- model
- unit
- answer
- check

Sequences: Worked Example

Worked example for Sequences: Continue the pattern 6, 12, 18, 24, __, __.

1. Compare terms: 12 - 6 = 6, 18 - 12 = 6.
2. Rule: add 6 each time.
3. 24 + 6 = 30, 30 + 6 = 36.

Answer: 30, 36.

Sequences: Vocabulary And Study Skill

Use these words and study moves before you practise Sequences. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Sequences
- method
- model
- unit
- answer
- check

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Sequences without a clear example, method, or evidence.

Sequences: Guided Activity

Guided activity for Sequences:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Sequences: Practice And Correction

Practice for Sequences:

Main outcome: understand, manipulate and use arithmetic, geometric sequences.

1. Continue the pattern 5, 10, 15, 20, __, __.
2. Write the rule for the pattern.
3. Create a similar pattern that increases by 4 each time.

Answer check:
Answers: 25, 30. Rule: add 5 each time.

Correction check:
- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to sequences and solve it.

Sequences: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Sequences.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- method is clear
- working is shown step by step
- answer is checked in context

Sequences: Outcome Check

Outcome check for Sequences: Understand, manipulate and use arithmetic, geometric sequences.

1. Continue 7, 14, 21, 28, __, __.
2. Write the rule.
3. Create one new pattern with the same rule.

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Sequences: Outcome Check

Outcome check for Sequences: Group investigation : If the bankrates increase or decrease unexpectedly, learners discuss or investigate how in the next n- years:.

1. Continue 7, 14, 21, 28, __, __.
2. Write the rule.
3. Create one new pattern with the same rule.

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Logarithmic and exponential equations: Lesson Opener

Learning focus: Logarithmic and exponential equations

Country link: a Rwanda school, market, farming, building, transport, or data problem for Logarithmic and exponential equations.

By the end of this topic, you should be able to:
- solve equations involving logarithms or exponentials and apply them to model and solve related problems

Inquiry questions:
- How can logarithmic and exponential equations help you solve a real problem in Subsidiary Mathematics?

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Logarithmic and exponential equations: Learn

Logarithmic and exponential equations teaches a mathematical idea by using a clear model, a labelled representation, step-by-step working, and a final check.

Key words:
- Logarithmic and exponential equations
- Logarithmic
- exponential
- equations
- method
- model
- unit
- answer

Logarithmic and exponential equations: Worked Example

Worked example for Logarithmic and exponential equations: Read the task, choose a representation, solve one step at a time, and check the answer in context.

Model:
1. Name what is known.
2. Choose a diagram, table, equation, or model.
3. Complete the calculation or reasoning.
4. Check the result with estimation or inverse reasoning.

Use a Rwanda classroom, market, garden, transport, or home example.

Logarithmic and exponential equations: Vocabulary And Study Skill

Use these words and study moves before you practise Logarithmic and exponential equations. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Logarithmic and exponential equations
- Logarithmic
- exponential
- equations
- method
- model
- unit
- answer

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Logarithmic and exponential equations without a clear example, method, or evidence.

Logarithmic and exponential equations: Guided Activity

Guided activity for Logarithmic and exponential equations:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Logarithmic and exponential equations: Practice And Correction

Practice for Logarithmic and exponential equations:

Main outcome: solve equations involving logarithms or exponentials and apply them to model and solve related problems.

1. Solve one logarithmic and exponential equations question using labelled steps.
2. Show your method with a drawing, table, number sentence, or diagram.
3. Check your answer and write one sentence explaining it.

Answer check:
Your answer should show a clear model, correct method, correct labels, and a final sentence.

Correction check:
- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to logarithmic and exponential equations and solve it.

Logarithmic and exponential equations: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Logarithmic and exponential equations.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- method is clear
- working is shown step by step
- answer is checked in context

Logarithmic and exponential equations: Outcome Check

Outcome check for Logarithmic and exponential equations: Solve equations involving logarithms or exponentials and apply them to model and solve related problems.

1. Write one clear example from the topic.
2. Solve it with labelled steps.
3. Check the answer in context.

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Trigonometric and inverse trigonometric functions: Lesson Opener

Learning focus: Trigonometric and inverse trigonometric functions

Country link: a Rwanda school, market, farming, building, transport, or data problem for Trigonometric and inverse trigonometric functions.

By the end of this topic, you should be able to:
- apply theorems of limits and formulas of derivatives to solve problems including trigonometric functions, optimization, motion,
- domain and range of a function
- parity of a function (odd or even)

Inquiry questions:
- How can trigonometric and inverse trigonometric functions help you solve a real problem in Subsidiary Mathematics?

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Trigonometric and inverse trigonometric functions: Learn

Trigonometric and inverse trigonometric functions teaches a mathematical idea by using a clear model, a labelled representation, step-by-step working, and a final check.

Key words:
- Trigonometric and inverse trigonometric functions
- Trigonometric
- inverse
- trigonometric
- functions
- method
- model
- unit

Trigonometric and inverse trigonometric functions: Worked Example

Worked example for Trigonometric and inverse trigonometric functions: Read the task, choose a representation, solve one step at a time, and check the answer in context.

Model:
1. Name what is known.
2. Choose a diagram, table, equation, or model.
3. Complete the calculation or reasoning.
4. Check the result with estimation or inverse reasoning.

Use a Rwanda classroom, market, garden, transport, or home example.

Trigonometric and inverse trigonometric functions: Vocabulary And Study Skill

Use these words and study moves before you practise Trigonometric and inverse trigonometric functions. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Trigonometric and inverse trigonometric functions
- Trigonometric
- inverse
- trigonometric
- functions
- method
- model
- unit

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Trigonometric and inverse trigonometric functions without a clear example, method, or evidence.

Trigonometric and inverse trigonometric functions: Guided Activity

Guided activity for Trigonometric and inverse trigonometric functions:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Trigonometric and inverse trigonometric functions: Practice And Correction

Practice for Trigonometric and inverse trigonometric functions:

Main outcome: apply theorems of limits and formulas of derivatives to solve problems including trigonometric functions, optimization, motion,.

1. Solve one trigonometric and inverse trigonometric functions question using labelled steps.
2. Show your method with a drawing, table, number sentence, or diagram.
3. Check your answer and write one sentence explaining it.

Answer check:
Your answer should show a clear model, correct method, correct labels, and a final sentence.

Correction check:
- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to trigonometric and inverse trigonometric functions and solve it.

Trigonometric and inverse trigonometric functions: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Trigonometric and inverse trigonometric functions.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- method is clear
- working is shown step by step
- answer is checked in context

Trigonometric and inverse trigonometric functions: Outcome Check

Outcome check for Trigonometric and inverse trigonometric functions: Apply theorems of limits and formulas of derivatives to solve problems including trigonometric functions, optimization, motion,.

1. Write one clear example from the topic.
2. Solve it with labelled steps.
3. Check the answer in context.

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Trigonometric and inverse trigonometric functions: Outcome Check

Outcome check for Trigonometric and inverse trigonometric functions: Domain and range of a function.

1. Write one clear example from the topic.
2. Solve it with labelled steps.
3. Check the answer in context.

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Vector space of real numbers: Lesson Opener

Learning focus: Vector space of real numbers

Country link: a Rwanda school, market, farming, building, transport, or data problem for Vector space of real numbers.

By the end of this topic, you should be able to:
- apply vectors of 3 to solve problems related to angles using the scalar product in 3

Inquiry questions:
- How can vector space of real numbers help you solve a real problem in Subsidiary Mathematics?

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Vector space of real numbers: Learn

Vector space of real numbers teaches how lines and angles describe shapes and directions.

Core idea:
- A line can be straight, parallel, perpendicular, horizontal, vertical, or slanting.
- An angle is a turn between two lines.
- Draw and label diagrams carefully before measuring or classifying.

Key words:
- Vector space of real numbers
- Vector
- space
- real
- numbers

Vector space of real numbers: Worked Example

Worked example for Vector space of real numbers: A shape has one right angle. How can you show it clearly?

1. Draw the two meeting lines.
2. Mark the square corner to show a right angle.
3. Label it 90 degrees.
4. Name a classroom example, such as a book corner.

Answer: A right angle is a quarter turn, 90 degrees.

Vector space of real numbers: Vocabulary And Study Skill

Use these words and study moves before you practise Vector space of real numbers. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Vector space of real numbers
- Vector
- space
- real
- numbers
- method
- model
- unit

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Vector space of real numbers without a clear example, method, or evidence.

Vector space of real numbers: Guided Activity

Guided activity for Vector space of real numbers:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Vector space of real numbers: Practice And Correction

Practice for Vector space of real numbers:

Main outcome: apply vectors of 3 to solve problems related to angles using the scalar product in 3.

1. Draw one right angle and label it 90 degrees.
2. Name one pair of parallel lines in the classroom.
3. Use a ruler to draw a straight line segment and label its endpoints.

Answer check:
Answer check: the right angle should show a square corner; parallel lines stay the same distance apart.

Correction check:
- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to vector space of real numbers and solve it.

Vector space of real numbers: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Vector space of real numbers.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- method is clear
- working is shown step by step
- answer is checked in context

Vector space of real numbers: Outcome Check

Outcome check for Vector space of real numbers: Apply vectors of 3 to solve problems related to angles using the scalar product in 3.

1. Draw and label one right angle.
2. Name one pair of parallel lines.
3. Explain how you know.

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Matrices and determinants of order: Lesson Opener

Learning focus: Matrices and determinants of order

Country link: a Rwanda school, market, farming, building, transport, or data problem for Matrices and determinants of order.

By the end of this topic, you should be able to:
- apply matrix and determinant of order 3 to solve related problems

Inquiry questions:
- How can matrices and determinants of order help you solve a real problem in Subsidiary Mathematics?

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Matrices and determinants of order: Learn

Matrices and determinants of order teaches a mathematical idea by using a clear model, a labelled representation, step-by-step working, and a final check.

Key words:
- Matrices and determinants of order
- Matrices
- determinants
- order
- method
- model
- unit
- answer

Matrices and determinants of order: Worked Example

Worked example for Matrices and determinants of order: Read the task, choose a representation, solve one step at a time, and check the answer in context.

Model:
1. Name what is known.
2. Choose a diagram, table, equation, or model.
3. Complete the calculation or reasoning.
4. Check the result with estimation or inverse reasoning.

Use a Rwanda classroom, market, garden, transport, or home example.

Matrices and determinants of order: Vocabulary And Study Skill

Use these words and study moves before you practise Matrices and determinants of order. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Matrices and determinants of order
- Matrices
- determinants
- order
- method
- model
- unit
- answer

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Matrices and determinants of order without a clear example, method, or evidence.

Matrices and determinants of order: Guided Activity

Guided activity for Matrices and determinants of order:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Matrices and determinants of order: Practice And Correction

Practice for Matrices and determinants of order:

Main outcome: apply matrix and determinant of order 3 to solve related problems.

1. Solve one matrices and determinants of order question using labelled steps.
2. Show your method with a drawing, table, number sentence, or diagram.
3. Check your answer and write one sentence explaining it.

Answer check:
Your answer should show a clear model, correct method, correct labels, and a final sentence.

Correction check:
- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to matrices and determinants of order and solve it.

Matrices and determinants of order: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Matrices and determinants of order.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- method is clear
- working is shown step by step
- answer is checked in context

Matrices and determinants of order: Outcome Check

Outcome check for Matrices and determinants of order: Apply matrix and determinant of order 3 to solve related problems.

1. Write one clear example from the topic.
2. Solve it with labelled steps.
3. Check the answer in context.

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Bivariate statistics: Lesson Opener

Learning focus: Bivariate statistics

Country link: a Rwanda school, market, farming, building, transport, or data problem for Bivariate statistics.

By the end of this topic, you should be able to:
- extend understanding, analysis and interpretation of bivariate data to correlation coefficients and regression lines

Inquiry questions:
- How can bivariate statistics help you solve a real problem in Subsidiary Mathematics?

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Bivariate statistics: Learn

Bivariate statistics teaches how to collect, organize, read, and explain information.

Core idea:
- Data must answer a clear question.
- Tables, tally charts, pictographs, and bar charts make data easier to read.
- A good conclusion says what the data shows.

Key words:
- Bivariate statistics
- Bivariate
- statistics
- method
- model
- unit
- answer
- check

Bivariate statistics: Worked Example

Worked example for Bivariate statistics: A class records favourite fruits: mango 8, banana 12, pineapple 6. Which fruit is most popular?

1. Read the table values.
2. Compare 8, 12, and 6.
3. 12 is the greatest number.

Answer: Banana is most popular.

Bivariate statistics: Vocabulary And Study Skill

Use these words and study moves before you practise Bivariate statistics. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Bivariate statistics
- Bivariate
- statistics
- method
- model
- unit
- answer
- check

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Bivariate statistics without a clear example, method, or evidence.

Bivariate statistics: Guided Activity

Guided activity for Bivariate statistics:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Bivariate statistics: Practice And Correction

Practice for Bivariate statistics:

Main outcome: extend understanding, analysis and interpretation of bivariate data to correlation coefficients and regression lines.

1. Ask 10 learners a simple question and make a tally table.
2. Draw a bar chart from the tally table.
3. Write one conclusion from the data.

Answer check:
Answer check: totals in the tally table and bar chart should match.

Correction check:
- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to bivariate statistics and solve it.

Bivariate statistics: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Bivariate statistics.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- method is clear
- working is shown step by step
- answer is checked in context

Bivariate statistics: Outcome Check

Outcome check for Bivariate statistics: Extend understanding, analysis and interpretation of bivariate data to correlation coefficients and regression lines.

1. Write one clear example from the topic.
2. Solve it with labelled steps.
3. Check the answer in context.

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Conditional probability and Bayes theorem: Lesson Opener

Learning focus: Conditional probability and Bayes theorem

Country link: a Rwanda school, market, farming, building, transport, or data problem for Conditional probability and Bayes theorem.

By the end of this topic, you should be able to:
- solve problems using Bayes theorem and use data to make decisions about likelihood and risk

Inquiry questions:
- How can conditional probability and bayes theorem help you solve a real problem in Subsidiary Mathematics?

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Conditional probability and Bayes theorem: Learn

Conditional probability and Bayes theorem teaches how likely an event is.

Core idea:
- Some events are certain, likely, unlikely, or impossible.
- A simple experiment can help you compare chances.
- Record results before making a conclusion.

Key words:
- Conditional probability and Bayes theorem
- Conditional
- probability
- Bayes
- theorem
- method
- model
- unit

Conditional probability and Bayes theorem: Worked Example

Worked example for Conditional probability and Bayes theorem: A bag has 4 red counters and 1 blue counter. Which colour is more likely to be picked?

1. Compare the number of counters.
2. Red has 4 chances; blue has 1 chance.
3. More red counters means red is more likely.

Answer: Red is more likely.

Conditional probability and Bayes theorem: Vocabulary And Study Skill

Use these words and study moves before you practise Conditional probability and Bayes theorem. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Conditional probability and Bayes theorem
- Conditional
- probability
- Bayes
- theorem
- method
- model
- unit

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Conditional probability and Bayes theorem without a clear example, method, or evidence.

Conditional probability and Bayes theorem: Guided Activity

Guided activity for Conditional probability and Bayes theorem:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Conditional probability and Bayes theorem: Practice And Correction

Practice for Conditional probability and Bayes theorem:

Main outcome: solve problems using Bayes theorem and use data to make decisions about likelihood and risk.

1. Write one certain event, one likely event, and one impossible event.
2. Toss a coin 10 times and record heads/tails.
3. Use your results to say what happened more often.

Answer check:
Answer check: certain means must happen; impossible means cannot happen.

Correction check:
- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to conditional probability and bayes theorem and solve it.

Conditional probability and Bayes theorem: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Conditional probability and Bayes theorem.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- method is clear
- working is shown step by step
- answer is checked in context

Conditional probability and Bayes theorem: Outcome Check

Outcome check for Conditional probability and Bayes theorem: Solve problems using Bayes theorem and use data to make decisions about likelihood and risk.

1. Write one clear example from the topic.
2. Solve it with labelled steps.
3. Check the answer in context.

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Trigonometric functions and equations: Review Clinic 1

Use this review clinic to strengthen Subsidiary Mathematics without copying earlier answers.

1. Choose one idea from Trigonometric functions and equations that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Sequences: Review Clinic 2

Use this review clinic to strengthen Subsidiary Mathematics without copying earlier answers.

1. Choose one idea from Sequences that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Logarithmic and exponential equations: Review Clinic 3

Use this review clinic to strengthen Subsidiary Mathematics without copying earlier answers.

1. Choose one idea from Logarithmic and exponential equations that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Trigonometric and inverse trigonometric functions: Review Clinic 4

Use this review clinic to strengthen Subsidiary Mathematics without copying earlier answers.

1. Choose one idea from Trigonometric and inverse trigonometric functions that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Glossary

Use this glossary to revise important words in Subsidiary Mathematics.

- Trigonometric functions and equations: Explain this term using an example from the book, your school, or your community.
- Sequences: Explain this term using an example from the book, your school, or your community.
- Logarithmic and exponential equations: Explain this term using an example from the book, your school, or your community.
- Trigonometric and inverse trigonometric functions: Explain this term using an example from the book, your school, or your community.
- Vector space of real numbers: Explain this term using an example from the book, your school, or your community.
- Matrices and determinants of order: Explain this term using an example from the book, your school, or your community.
- Bivariate statistics: Explain this term using an example from the book, your school, or your community.
- Conditional probability and Bayes theorem: Explain this term using an example from the book, your school, or your community.

Add five more words from your lessons and write your own meanings.

Answer And Teacher Notes

A strong Subsidiary Mathematics answer shows the model, method, units or labels, final answer, and a check. Use the worked examples and answer checks in each practice page to correct your work. When an answer is wrong, find the first step that changed the meaning and correct that step before trying a new question.

Final Project

Create a final project for Subsidiary Mathematics that shows what you can now do independently.

Your project must include:
1. A clear title.
2. A real-life problem, question, text, performance, investigation, design, or worked task.
3. Evidence from at least three lessons in this book.
4. A drawing, table, map, chart, model, dialogue, paragraph, performance plan, practical product, or worked solution.
5. A correction note showing one improvement after feedback.
6. A short reflection explaining what you learned, what was difficult, and what you will practise next.

Presentation check: keep the work neat, label important parts, use correct vocabulary, and be ready to explain your choices to a classmate or teacher.